MAHIR AFSER PAVEL

AI Researcher & Graduate Assistant Candidate

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Profile Summary

An AI researcher with a strong research interest in **medical image analysis** and **100+ citations** in **5 Q1** and solo Q2 journal publications, **4** of which are **SCIE** and **2 ESCI**, holding an **h-index** of **3** and an **i10-index** of **2**.

Research Interests

- Vision-Language Models (VLMs) & Large Language Models (LLMs)
- Deep Learning, Explainable Artificial Intelligence (XAI), & Edge AI
- Medical Image Analysis & Computational Clinical Triaging

Education

Degree	Major	Institution	CGPA	Dates Attended
Bachelor of Science	Computer Science and Engineering	North South University	***3.71****/4.00* (almost ***90%*** marks) | CSE Major (Capstone Design + Core + Elective Courses): ***3.91****/4.00* | Rank: ***38****/595*	2019 – 2024

B.Sc. Thesis: *"Non-small cell lung cancer detection through knowledge distillation approach with teaching assistant"*

- **Supervisor:** Dr. Riasat Khan (Associate Professor, Department of ECE).
- **Summary:** Implemented a three-stage knowledge distillation architecture for lung cancer classification on the *NSCLC-Radiomics dataset, which* achieved 94.53% accuracy and was later deployed on a user-friendly web application.

Scholarly Publications

Peer-Reviewed Journal Publications

- ***Multi-stage knowledge distillation with layer fusion-based deep learning approach for skin cancer classification***.
  - ✔ ***Authors: Mahir Afser Pavel****, Ramisa Asad, Goh Kah Ong Michael, Md Ikramuzzaman, Murad Mustakim, Riasat Khan.*
  - ✔ ***Venue****: Scientific Reports (Nature Portfolio), 2025. DOI: 10.1038/s41598-025-23403-2 [Code]*
- ***Non-small cell lung cancer detection through knowledge distillation approach with teaching assistant***
  - ✔ ***Authors: Mahir Afser Pavel****, Rafiul Islam, Shoyeb Bin Babor, Riaz Mehadi, Riasat Khan*
  - ✔ ***Venue****: PLOS ONE, 2024. DOI: 10.1371/journal.pone.0306441 [Code]*
- ***Error-Tolerant Multimodal Vision-Language Models for Endodontic Triaging: A Cross-Sectional Study***.
  - ✔ ***Authors: Mahir Afser Pavel*** *(Co-First Author), Md Fahim Shahoriar Titu, Afifa Zain Apurba, Saif Ahmed, Shafin Rahman, James Dudley, Taseef Hasan Farook*
  - ✔ ***Venue****: International Journal of Dentistry (Wiley), 2026. DOI: 10.1155/ijod/4148741. [Code]*
- ***Fusion of Image Filtering and Knowledge-Distilled YOLO Models for Root Canal Failure Diagnosis***
  - ✔ ***Authors:*** *Afifa Zain Apurba, Md Fahim Shahoriar Titu,* ***Mahir Afser Pavel****, Intisar Tahmid Naheen, Riasat Khan*
  - ✔ ***Venue:*** *IEEE Access, 2025. DOI: 10.1109/ACCESS.2025.3560998. [Code]*
- ***Real-time fire detection: Integrating lightweight deep learning models on drones with edge computing***
  - ✔ ***Authors:*** *Md Fahim Shahoriar Titu,* ***Mahir Afser Pavel****, Goh Kah Ong Michael, Hisham Babar, Umama Aman, Riasat Khan.*
  - ✔ ***Venue****: Drones (MDPI), 2024. DOI: 10.3390/drones8090483 [Code]*
- ***Hybrid ViT-RetinaNet with Explainable Ensemble Learning for Fine-Grained Vehicle Damage Classification***
  - ✔ ***Authors****: Ananya Saha,* ***Mahir Afser Pavel****, Md Fahim Shahoriar Titu, Afifa Zain Apurba, Riasat Khan.*
  - ✔ ***Venue:*** *Vehicles (MDPI), 2025. DOI: 10.3390/vehicles7030089. [Code]*

Ongoing Research Publications

- ***Federated Multi-Component Framework for Brain Tumor Diagnosis with Generative Augmentation and Explainable AI***
  - ***Authors: Mahir Afser Pavel, Nafiz Fahad, Md Tanzib Hosain, Md. Kishor Morol, Tze Hui Liew***
  - ***Status: Under Review at Array (array), 2026***

- ***CLKD-MED: A novel cross-lingual knowledge distillation framework for multilingual clinical outcome prediction***
  - ***Authors: Mahir Afser Pavel, Rafiul Islam, Mohammad Junayed Hasan, and M. R. C. Mahdy***
  - ***Status: Under Review at Information Processing & Management (IPM), 2026***
- ***SLGRA-Net: Structure-Aware Latent Graph Reasoning for 3D Brain Tumor Segmentation***
  - ***Authors: Mahir Afser Pavel, Nafiz Fahad, Md Tanzib Hosain, Md. Kishor Morol, Tze Hui Liew***
  - ***Status: Under Review at Biomedical Signal Processing and Control (BSPC), 2026***
- ***Trustworthy Clinical AI for Diabetic Retinopathy Grading via Hybrid Foundation Models, Calibration-Aware Uncertainty Learning, and Multi-Criteria Clinical Referral***
  - ***In Progress, 2026***
- ***S³-Net: Self-Stabilizing, Self-Regulating, and Self-Optimizing Neural Networks via Universal Stability Learning***
  - ***In Progress, 2026***
- ***FULCRUM: Option-Contrastive Frame Selection for Long-Video Understanding***
  - ***In Progress, 2026***
- ***ACP-KD: Adaptive Clinical Prototype Knowledge Distillation with Dynamic Feature Fusion and Confidence-Aware Multi-Objective Learning for Lightweight Skin Lesion Classification***
  - ***In Progress, 2026***

Professional Experience

Research Assistant | ELITE Research Lab LLC (Dhaka, Bangladesh) (Aug 2025 – Present)

- Federated Medical Architectures: Engineered "Fed-GAN-DE-CL," a federated generative adversarial network utilizing dual-encoder contrastive learning, to enable decentralized brain tumor classification while preserving strict patient data privacy.
- Topological Segmentation: Programmed "SLGRA-Net," incorporating structure-aware latent graph reasoning to enhance boundary definitions in 3D brain tumor MRI segmentations, resolving topological boundary ambiguity.
- Explainable AI Integration: Built an uncertainty-aware diabetic retinopathy grading framework, deploying Monte Carlo Dropout to estimate predictive variance and Grad-CAM for diagnostic explanations.

Research Assistant | United International University (Dhaka, Bangladesh) (May 2025 – Jun 2025)

- Pipeline Optimization: Designed and executed end-to-end data preprocessing, feature engineering, and experimental training pipelines for deep learning and natural language processing (NLP) models.
- Literature Synthesis: Authored thorough literature reviews on emerging multi-agent paradigms and co-drafted research proposals.

Research Assistant | North South University (Dhaka, Bangladesh) (May 2024 – Jul 2024)

- **Edge Computer Vision**: Co-developed real-time drone-based fire detection frameworks, compressing deep models using autodistill-based knowledge distillation to deploy YOLOv8n onto Raspberry Pi 5 edge devices. Improved target detection accuracy by 10-20% under dynamic outdoor constraints.

Research Intern | Mahdy Research Academy (Remote / Online) (Jan 2024 – Mar 2025)

- **Clinical Translation Frameworks**: Developed "CLKD-MED," a cross-lingual knowledge distillation framework, mapping semantic clinical representations across languages to predict patient outcomes from multilingual electronic health records.

Teaching & Research Experience

- Conducted primary research as a key researcher within the **ELITE Research Lab LLC** and **Mahdy Research Academy**, driving the design and execution of deep learning, computer vision, and clinical NLP pipelines.
- Served as an active peer reviewer for **IEEE Access**, providing rigorous manuscript evaluations for submissions focusing on edge-deployed dual-agent systems and training-free zero-shot action recognition frameworks.
- Co-led hands-on academic sessions and supervised laboratory modules for Analog Electronics and Senior Design at North South University, directly mentoring student groups on circuit mapping, hardware-software co-design, and research publication drafts under faculty supervision.

Standardized Examination Scores

IELTS Academic (2026)	Listening: **7.0**	Reading: **7.0**	Writing: **6.5**	Speaking: **6.0**

Academic Projects & Research Focus

- **Quantization-Aware Edge AI:** Evaluating and optimizing lightweight deep learning models (YOLOv5, YOLOv8) and multimodal vision-language models (BLIP, CLIP, Florence-2, PaliGemma) using Low-Rank Adaptation (LoRA) and quantization-aware training to minimize memory consumption by up to 87.5% for resource-constrained clinical settings.
- **Oncology Image Analytics & Distillation:** Refining multi-stage teacher-assistant-student distillation pipelines using Vision Transformers (ViT) and ResNet152v2 on clinical radiography datasets (HAM10000, NSCLC CT scans), enhancing diagnostic sensitivity while preserving model parameters.
- **Privacy-Preserving Biomedical Machine Learning:** Architecting federated generative adversarial networks and contrastive learning methodologies to facilitate secure, collaborative model training on decentralized, multi-institutional clinical databases.

Technical & Research Skills

Category	Specific Tools and Proficiency Level
Programming Languages	Python (Advanced), SQL, C++, Java, Bash Scripting
AI/ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Detectron2
Generative AI & VLMs	LangChain, LLaMA, RAG, Low-Rank Adaptation (LoRA), QAT, BLIP, CLIP, Florence-2, PaliGemma
Edge AI & Hardware	Raspberry Pi 5, DJI F450 custom drone, Pixhawk PX4, Git, GitHub, LaTeX, Overleaf
Databases & Analysis	SQL, MySQL, NumPy, Pandas, Matplotlib, SPSS, Microsoft Excel

Leadership & Extracurricular Activities

- Research Team Coordinator (2023 – 2024): Supervised a team of 5 research assistants across senior design and directed research projects, coordinating data pipelines and draft submissions that culminated in 5 peer-reviewed journal articles under faculty mentorship.
- Manuscript Reviewer (2026): Provided critical analytical reviews of manuscripts detailing edge AI architectures and zero-shot learning frameworks, maintaining high standards for the journal's methodology criteria.
- Academic Peer Mentor (2022–2023): Guided junior engineering students in laboratory environments, resolving hardware-software interfacing issues, and assisting in scientific writing workshops

Honors & Awards

- **Magna Cum Laude (December 2024)**: Conferred by North South University for outstanding academic excellence in the B.Sc. in Computer Science and Engineering program, acknowledging placement within the top 7% in the cohort.
- **Academic Merit Scholarship**: Recipient of multiple tuition waivers (10%–20%) from North South University based on high semester grade point averages (semester GPA up to 4.00).
- **Peer Reviewer Recognition (March 2026):** Recognized for critical scientific reviews of edge-deployed systems and action recognition manuscripts in IEEE Access, Heliyon, and Frontiers in Computer Science.

References

Dr. Riasat Khan, *Associate Professor, Dept. of ECE,* North South University Email: riasat.khan@northsouth.edu	**Dr. Shafin Rahman,** *Associate Professor, Dept. of ECE,* North South University Email: shafin.rahman@northsouth.edu	**Md. Kishor Morol,** *Research Scientist & Founder,* ELITE Research Lab LLC Email: kishor@aiub.edu